Detailed Connections Between Arduino, 7447, and Two 7474 ICs

Aim

To design and implement a BCD (Binary Coded Decimal) counter circuit using Arduino Uno, two 7474 Dual D Flip-Flop ICs, a 7447 BCD-to-Seven-Segment Decoder, and a Common Anode Seven Segment Display that counts from 0 to 9 and resets back to 0.

1. Components Used

Component
Arduino Uno
7447 BCD to Seven Segment Decoder
7474 IC #1 (Dual D Flip-Flop)
7474 IC #2 (Dual D Flip-Flop)
Common Anode Seven Segment Display
Breadboard
Jumper Wires

2. Purpose of Each Component

Component	Function
Arduino	Generates clock and incrementing signals
7474 ICs	Store binary count states
7447	Converts BCD input into seven-segment signals
Seven Segment Display	Displays decimal digits

3. Arduino Connections

The Arduino provides:
• Clock signal
• Incrementing logic
• Input/output control

Arduino Pin	Signal
D2	A
D3	B
D4	C
D5	D
D6	W
D7	X
D8	Y
D9	Z

D13	Clock

4. First 7474 IC Connections

A 7474 IC contains two D flip-flops.

7474 Pin Diagram Important Pins

Pin	Function
1	CLR1
2	D1
3	CLK1
4	PRE1
5	Q1
6	Q■1
7	GND
8	Q■2
9	Q2
10	PRE2
11	CLK2
12	D2
13	CLR2
14	VCC

Power Connections

7474 Pin	Connection
Pin 14	+5V
Pin 7	GND

Arduino to First 7474 Connections

Arduino Pin	Signal	7474 Pin
D13	Clock	Pin 3 (CLK1)
D13	Clock	Pin 11 (CLK2)
D6	W	Pin 5 (Q1)
D7	X	Pin 9 (Q2)
D2	A	Pin 2 (D1)
D3	B	Pin 12 (D2)

Control Pins

The preset and clear pins are active LOW. Keep them HIGH for normal operation.

7474 Pin	Connection
Pin 1 (CLR1)	+5V
Pin 4 (PRE1)	+5V
Pin 10 (PRE2)	+5V
Pin 13 (CLR2)	+5V

5. Second 7474 IC Connections

The second 7474 stores the remaining two bits.

Power Connections

7474 Pin	Connection
Pin 14	+5V
Pin 7	GND

Arduino to Second 7474 Connections

Arduino Pin	Signal	7474 Pin
D13	Clock	Pin 3 (CLK1)
D13	Clock	Pin 11 (CLK2)
D8	Y	Pin 5 (Q1)
D9	Z	Pin 9 (Q2)
D4	C	Pin 2 (D1)
D5	D	Pin 12 (D2)

Control Pins

7474 Pin	Connection
Pin 1 (CLR1)	+5V
Pin 4 (PRE1)	+5V
Pin 10 (PRE2)	+5V
Pin 13 (CLR2)	+5V

6. Connections Between Arduino and 7447 Decoder

The 7447 receives BCD inputs.

Arduino Signal	7447 Pin
A	Pin 7

B	Pin 1
C	Pin 2
D	Pin 6

7. Power Connections for 7447

7447 Pin	Connection
Pin 16	+5V
Pin 8	GND

8. 7447 to Seven Segment Display Connections

The outputs of 7447 are connected to the display segments.

7447 Output	Seven Segment
a	Segment a
b	Segment b
c	Segment c
d	Segment d
e	Segment e
f	Segment f
g	Segment g

9. Seven Segment Common Anode Connection

Since the display is Common Anode:

Display Pin	Connection
Common Anode	+5V

10. Overall Working

1. Arduino generates clock pulses.
2. Clock pulses trigger the 7474 flip-flops.
3. Flip-flops store binary states.
4. Outputs A, B, C, D form a BCD number.
5. 7447 converts BCD into seven-segment control signals.
6. Display shows decimal digits from 0 to 9.
7. Feedback signals W, X, Y, Z help implement incrementing logic.

11. Data Flow of the System

Arduino Clock
↓
7474 Flip-Flops
↓
Binary Count (A B C D)

↓
7447 Decoder
↓
Seven Segment Display
↓
Displayed Decimal Number

12. Important Notes

• Use a Common Anode display only.
• Keep all unused inputs HIGH.
• Ensure common GND between Arduino and all ICs.
• The 7447 works with BCD input only.
• The clock signal should be slow enough to observe counting.